HSE 103 Final Project: Submission 2

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Data Import

```r
# Read in covariate data
covarData <- read.csv(file = "GSE85307_covar-1.csv")

# Read in gene expression data
geneExpression <- read.csv(file = "GSE85307_geneExpression.csv",
                           row.names = 1)

# Read in gene name linker
geneNames <- read.csv(file = "GSE85307_geneNames.csv",
                      row.names = 1)

# Check that sample IDs match
all(covarData$geoID %in% colnames(geneExpression))
```

```
## [1] TRUE
```

```r
all(colnames(geneExpression) %in% covarData$geoID)
```

```
## [1] TRUE
```

Selected Genes

For Submission 2, I retained **TBL1X** from Submission 1 and added **KDM1B** and **TMEM176B**.

```r
# Store selected genes
selectedGenes <- c("TBL1X", "KDM1B", "TMEM176B")

# Identify probe IDs using R
geneProbeMap <- geneNames[
  geneNames$GENE %in% selectedGenes,
  ,
  drop = FALSE
]

# Save probe IDs from row names
geneProbeMap$PROBEID <- rownames(geneProbeMap)

# Put genes in selected order
geneProbeMap <- geneProbeMap[
  match(selectedGenes, geneProbeMap$GENE),
  ,
  drop = FALSE
]

# Check probe IDs
geneProbeMap
```

```
##             GENE PROBEID
## 8165911    TBL1X 8165911
## 8117081    KDM1B 8117081
## 8143790 TMEM176B 8143790
```

```r
# Verify that all selected genes and probe IDs were found
all(selectedGenes %in% geneProbeMap$GENE)
```

```
## [1] TRUE
```

```r
all(geneProbeMap$PROBEID %in% rownames(geneExpression))
```

```
## [1] TRUE
```

Data Wrangling

```r
# Create project data frame
projectData <- covarData

# Remove text from BMI values
projectData$mat_BMI <- sub("bmi_kg/m2: ",
                           "",
                           projectData$mat_BMI)

# Convert BMI to numeric
projectData$mat_BMI <- as.numeric(projectData$mat_BMI)

# Calculate median BMI
medianBMI <- median(projectData$mat_BMI,
                    na.rm = TRUE)

# Create categorical BMI variable
# Labels were changed based on Submission 1 feedback
projectData$BMIgroup <- cut(
  projectData$mat_BMI,
  breaks = c(-Inf, medianBMI, Inf),
  labels = c("Below Median BMI",
             "Above Median BMI")
)

# Remove text from vitamin D values
projectData$mat_vitD <- sub(
  "vitamind_25\\(0h\\)_ng/ml_whole_blood_baseline: ",
  "",
  projectData$mat_vitD
)

# Convert vitamin D to numeric
projectData$mat_vitD <- as.numeric(projectData$mat_vitD)

# Remove text from pregnancy outcome values
projectData$preg_preeclampsia <- sub(
  "pregnancy_condition: ",
  "",
  projectData$preg_preeclampsia
)

# Set pregnancy outcome order
projectData$preg_preeclampsia <- factor(
  projectData$preg_preeclampsia,
  levels = c("Normal", "Preeclampsia")
)
```

Add Selected Gene Expression

```r
# Select expression rows using the probe IDs identified by R
selectedExpression <- geneExpression[
  geneProbeMap$PROBEID,
  projectData$geoID,
  drop = FALSE
]

# Replace probe IDs with gene names
rownames(selectedExpression) <- geneProbeMap$GENE

# Transpose so genes become columns
selectedExpression <- as.data.frame(t(selectedExpression))

# Combine expression data with covariate data
analysisData <- cbind(projectData,
                      selectedExpression)

# Check selected gene values
summary(analysisData[, selectedGenes])
```

```
##      TBL1X           KDM1B          TMEM176B
##  Min.   :8.283   Min.   :7.547   Min.   :7.373
##  1st Qu.:8.955   1st Qu.:8.191   1st Qu.:7.842
##  Median :9.194   Median :8.393   Median :8.278
##  Mean   :9.165   Mean   :8.374   Mean   :8.416
##  3rd Qu.:9.421   3rd Qu.:8.569   3rd Qu.:8.933
##  Max.   :9.862   Max.   :8.955   Max.   :9.959
```

```r
# Check categorical variables
table(analysisData$BMIgroup)
```

```
##
## Below Median BMI Above Median BMI
##               79               78
```

```r
table(analysisData$preg_preeclampsia)
```

```
##
##       Normal Preeclampsia
##          110           47
```

Plotting Function

The function below takes a data frame and a list of one or more gene names. It loops through each gene and creates the three plots required for the project.

```r
buildGenePlots <- function(data, geneList) {

  # Create empty list to store plots
  plotList <- list()

  # Loop through each gene
  for (gene in geneList) {

    # Make a temporary copy of the data
    tempData <- data

    # Save the current gene expression values in one column
    tempData$geneExpression <- tempData[, gene]

    # Histogram
    histPlot <- ggplot(tempData,
                       aes(x = geneExpression)) +
      geom_histogram(bins = 30) +
      labs(title = paste(gene,
                         "Gene Expression Distribution"),
           x = paste(gene,
                     "Gene Expression"),
           y = "Frequency") +
      theme_classic()

    # Scatterplot
    scatterPlot <- ggplot(tempData,
                          aes(x = mat_vitD,
                              y = geneExpression)) +
      geom_point() +
      geom_smooth(method = lm,
                  se = FALSE) +
      labs(title = paste("Maternal Vitamin D and",
                         gene,
                         "Expression"),
           x = "Maternal Vitamin D (ng/mL)",
           y = paste(gene,
                     "Gene Expression")) +
      theme_classic()

    # Boxplot
    boxPlot <- ggplot(tempData,
                      aes(x = preg_preeclampsia,
                          y = geneExpression,
                          fill = BMIgroup)) +
      geom_boxplot() +
      labs(title = paste(gene,
                         "Expression by Pregnancy Outcome and Maternal BMI"),
           x = "Pregnancy Outcome",
           y = paste(gene,
                     "Gene Expression"),
           fill = "Maternal BMI") +
      theme_classic()

    # Save plots to list
    plotList[[paste0(gene,
                     "_histogram")]] <- histPlot

    plotList[[paste0(gene,
                     "_scatterplot")]] <- scatterPlot

    plotList[[paste0(gene,
                     "_boxplot")]] <- boxPlot
  }

  return(plotList)
}
```

Generate the Nine Required Plots

```r
# Run function for all three genes
genePlots <- buildGenePlots(
  data = analysisData,
  geneList = selectedGenes
)

# Confirm that 9 plots were created
length(genePlots)
```

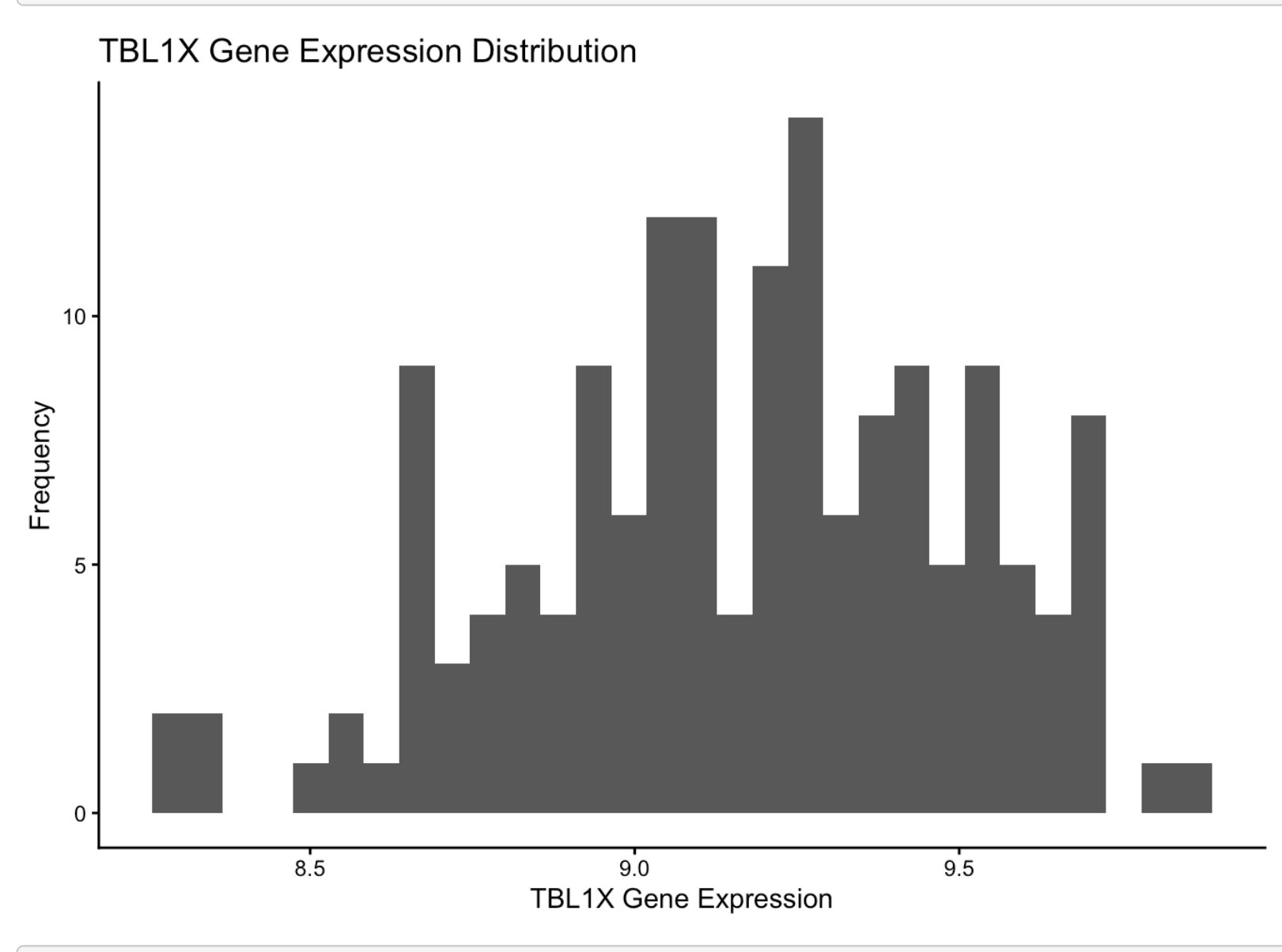
```
## [1] 9
```

```r
# Check plot names
names(genePlots)
```

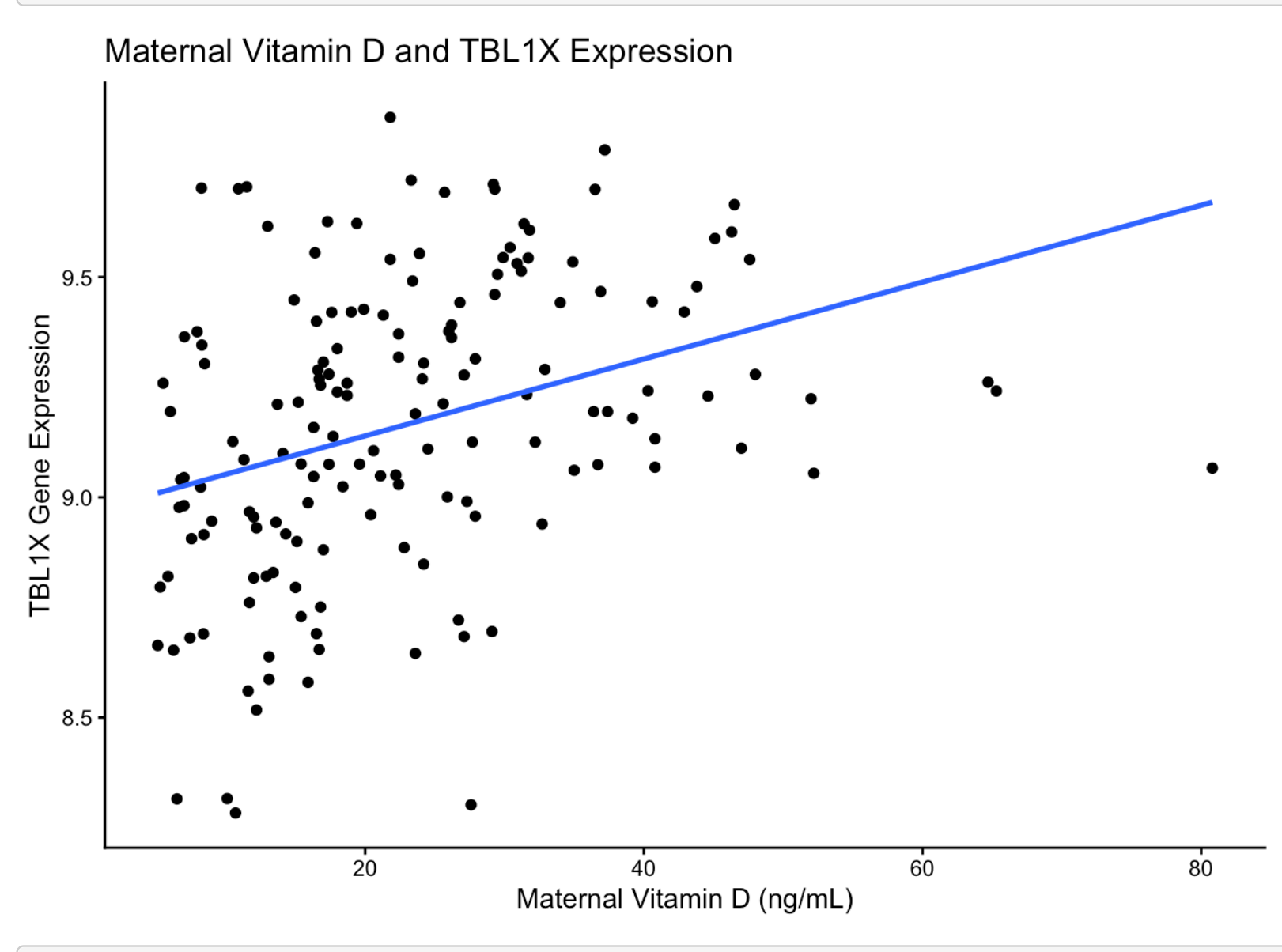
```
## [1] "TBL1X_histogram"      "TBL1X_scatterplot"    "TBL1X_boxplot"
## [4] "KDM1B_histogram"      "KDM1B_scatterplot"    "KDM1B_boxplot"
## [7] "TMEM176B_histogram"   "TMEM176B_scatterplot" "TMEM176B_boxplot"
```

TBL1X

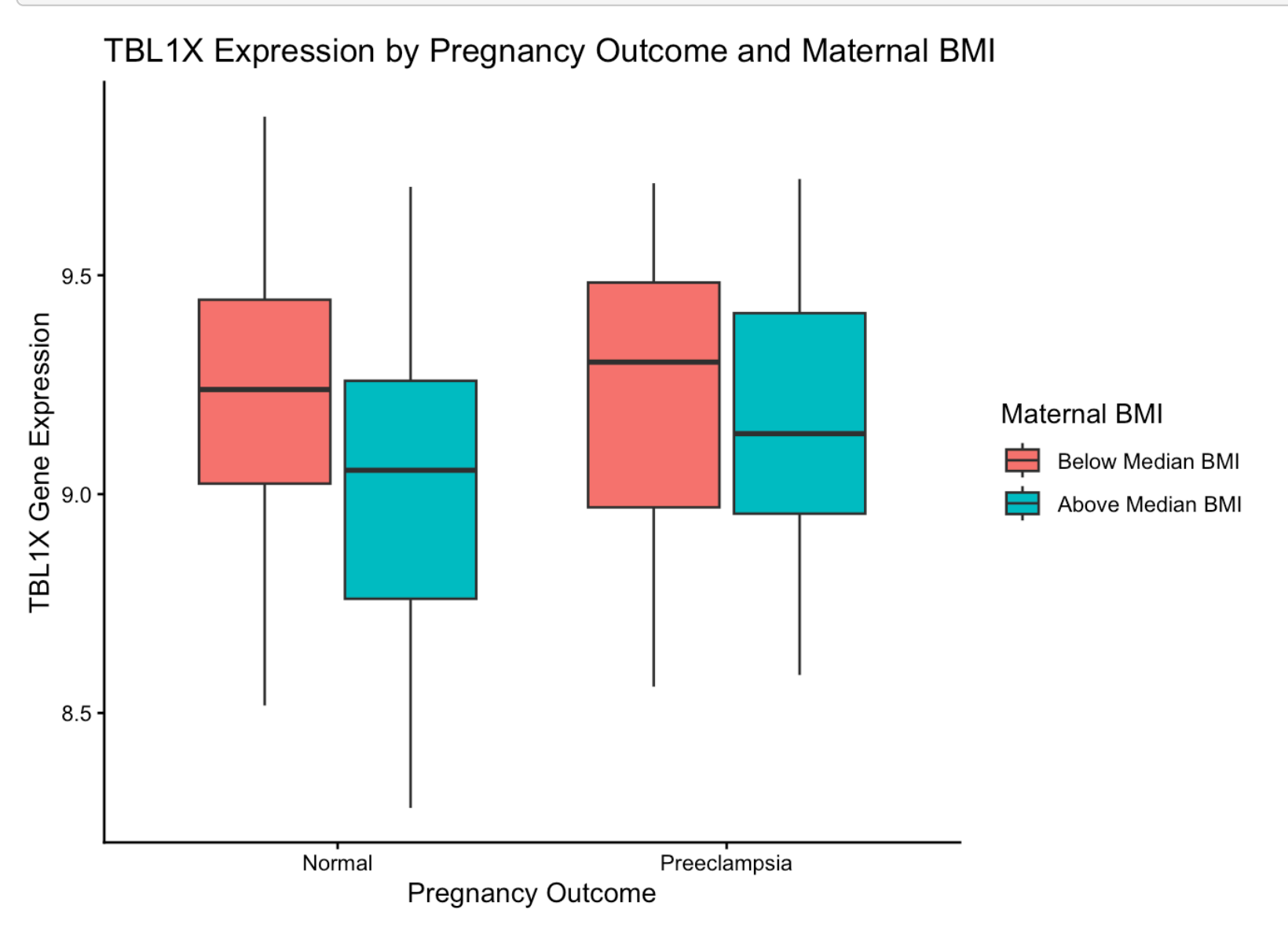
```r
genePlots[["TBL1X_histogram"]]
```



```r
genePlots[["TBL1X_scatterplot"]]
```

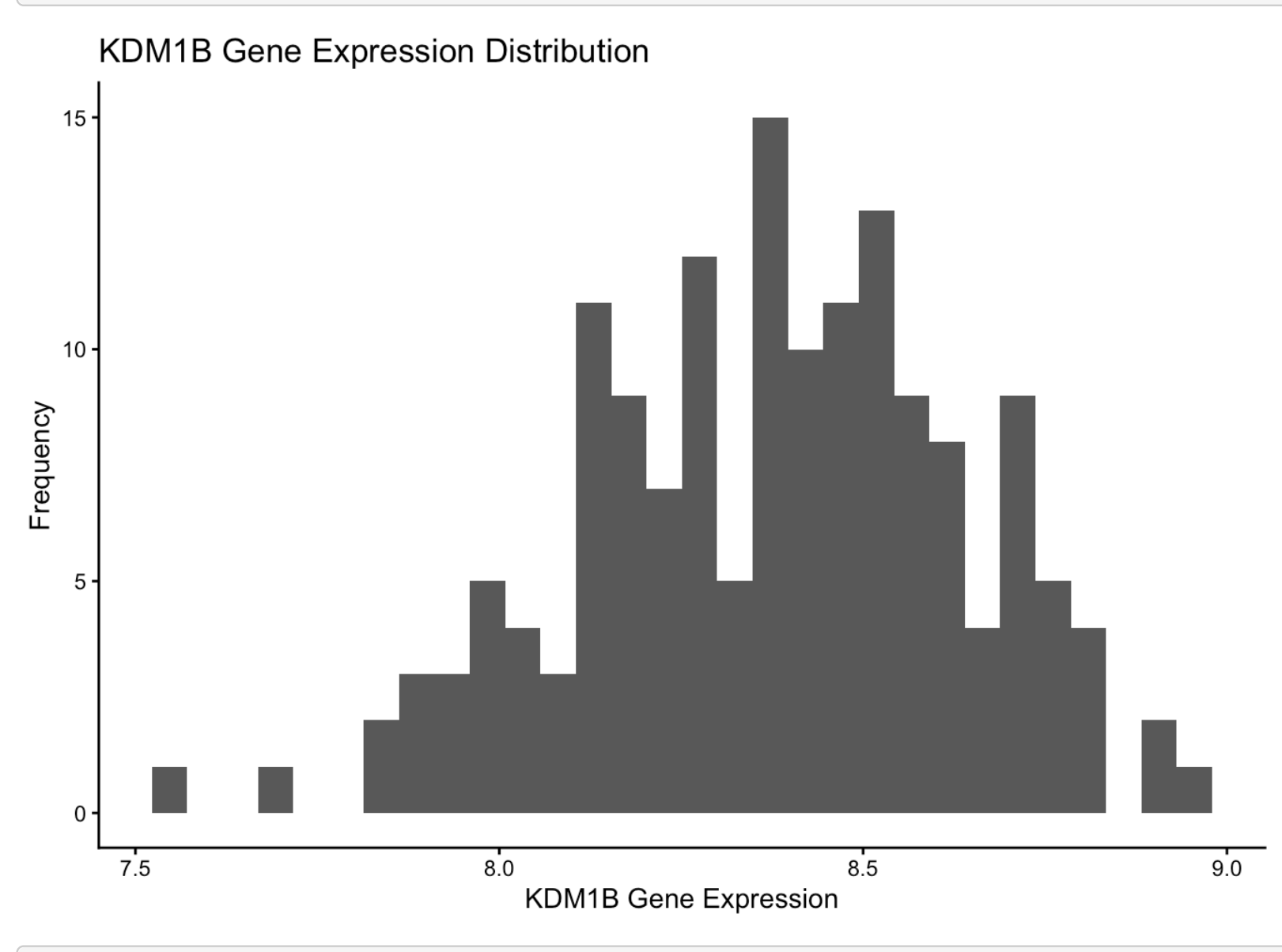


```r
genePlots[["TBL1X_boxplot"]]
```

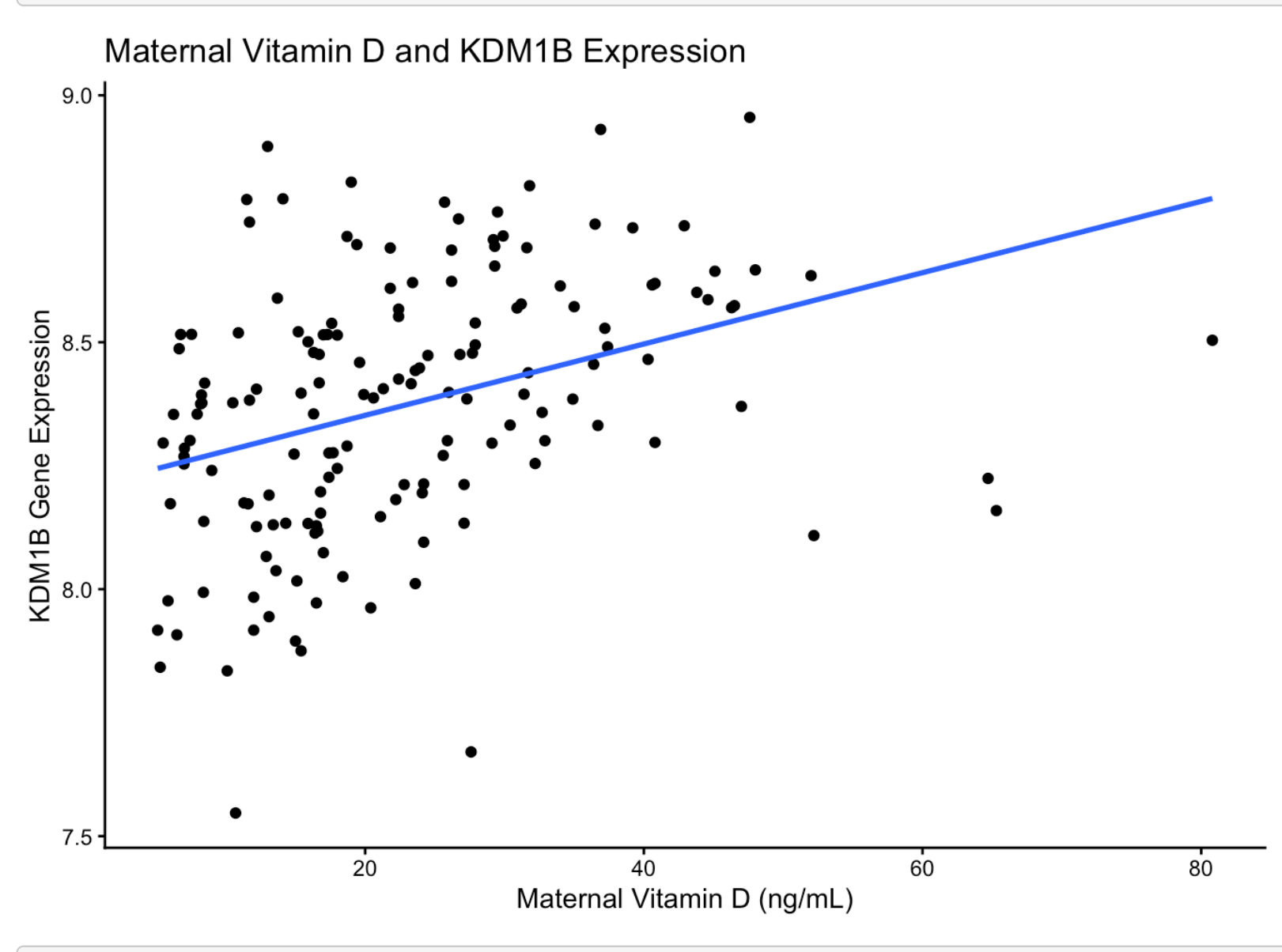


KDM1B

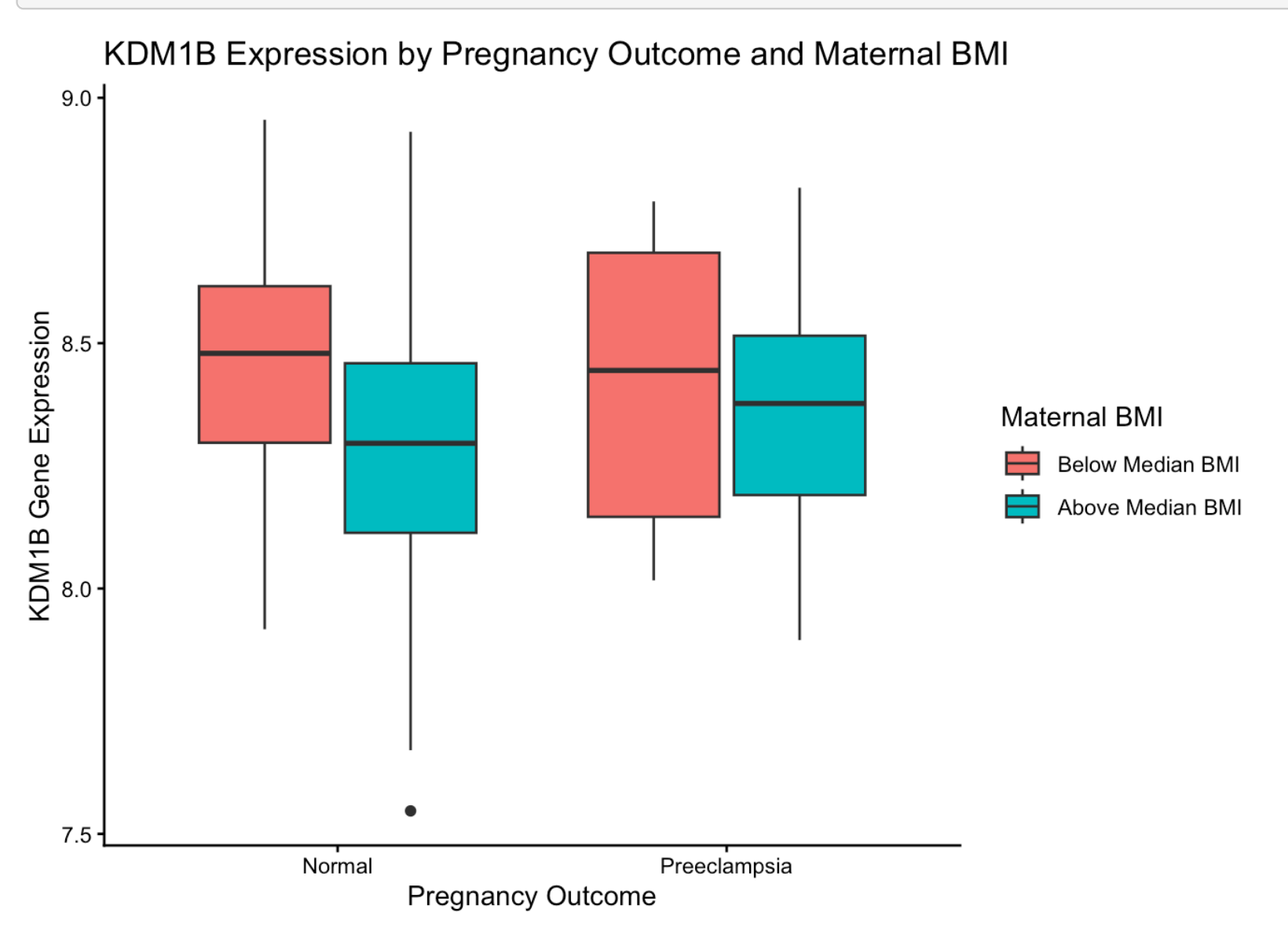
```r
genePlots[["KDM1B_histogram"]]
```



```r
genePlots[["KDM1B_scatterplot"]]
```

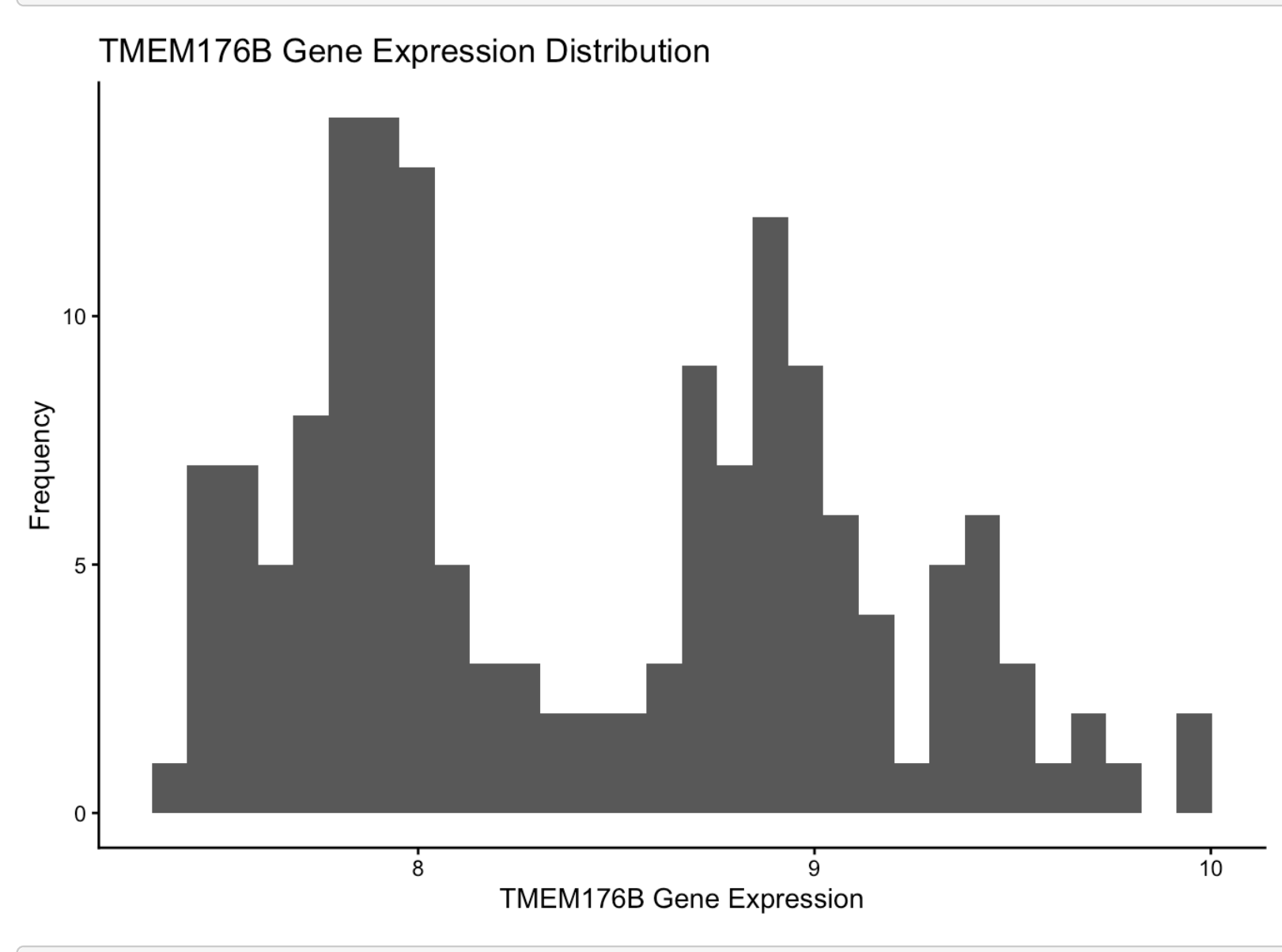


```r
genePlots[["KDM1B_boxplot"]]
```

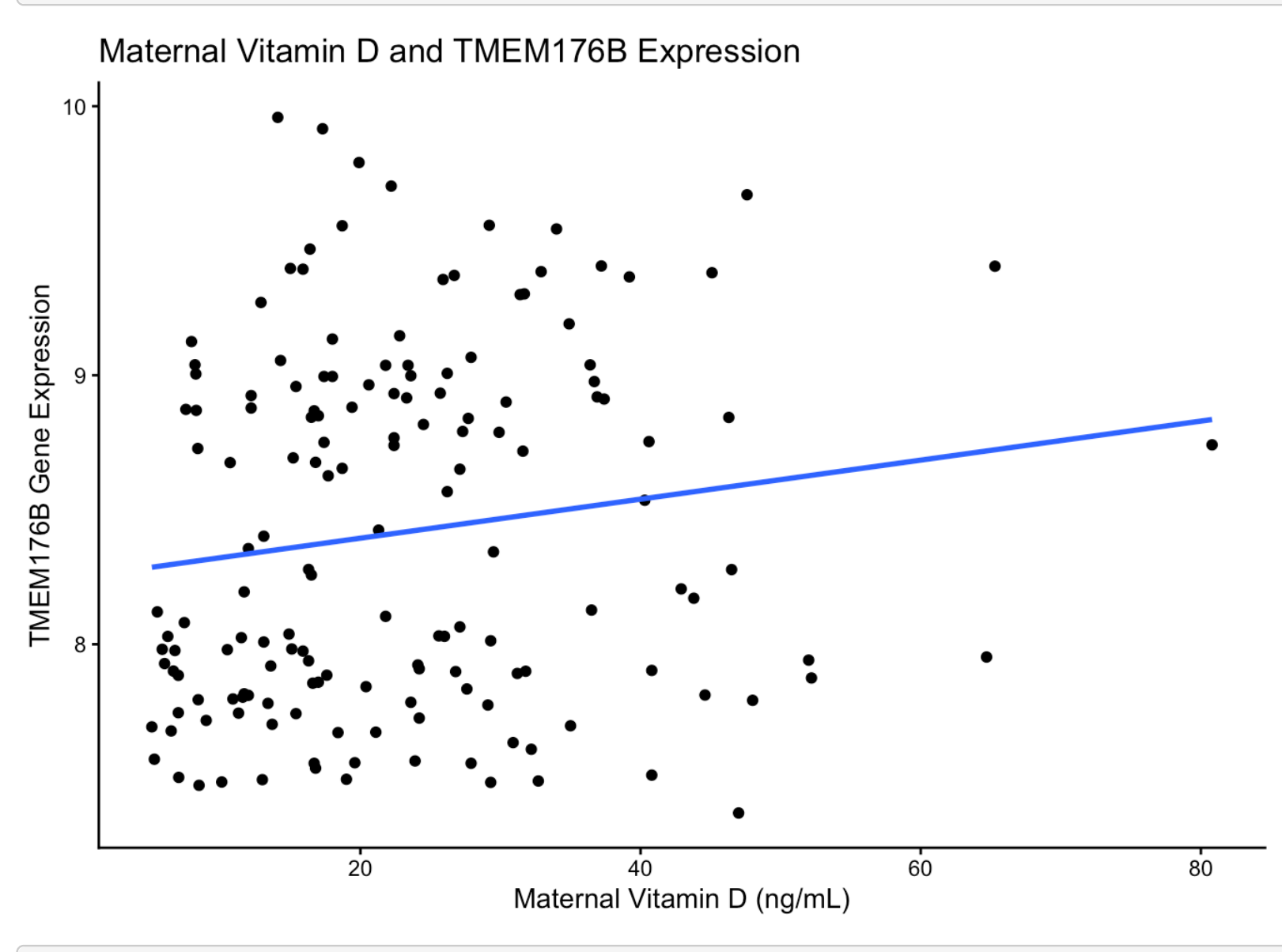


TMEM176B

```r
genePlots[["TMEM176B_histogram"]]
```



```r
genePlots[["TMEM176B_scatterplot"]]
```



```r
genePlots[["TMEM176B_boxplot"]]
```